Microsoft Entra ID – Identity & Access Management Lab

Azure AD Project | Edukamu Cloud and Cybersecurity Microdegree

Role-Based Access Model

For each team member, a dedicated Microsoft Entra ID user account is created using a corporate email format. All identities are centrally managed and assigned to role-based security groups to ensure least privilege access.

1. Team Lead

- Identity type: User account
- Assigned to: Leadership Group
- Access to project management system and reporting tools
- No global administrator rights to reduce security risk

2. Senior Programmer

- Identity type: User account
- Assigned to: Developers Group
- Access to source code repositories and cloud development resources
- No access to finance records or marketing data

3. Junior Programmer

- Identity type: User account
- Assigned to: Developers Group (restricted level)
- Limited access to development environments
- No direct access to production systems

4. Designer

- Identity type: User account
- Assigned to: Creative Group
- Access to design folders and shared marketing folders
- No access to code repositories or financial data

5. Tester

- Identity type: User account
- Assigned to: QA Group
- Access to testing environments and bug tracking systems
- No direct access to production database

6. Marketing Specialist

- Identity type: User account
- Assigned to: Marketing Group
- Access to marketing materials and analytics tools
- No access to source code or internal user database

Permissions will be assigned through security groups instead of directly to users. This simplifies governance and reduces the risk of configuration errors.

Authentication Methods

Passwordless Multi-Factor Authentication (MFA) will be implemented using Microsoft Entra ID and Microsoft Authenticator. This approach significantly improves security compared to traditional password-based authentication by reducing phishing and credential theft risks. When combined with Conditional Access policies, additional verification will only be required when necessary, such as during sign-ins from unfamiliar locations or devices. This ensures strong protection while maintaining usability for employees.

Access Management Strategy

Access to resources will be assigned using role-based security groups following the principle of least privilege. Sensitive systems such as user databases and finance records will be restricted to authorised roles only.

Production environments will be separated from development and testing environments to reduce operational and security risk. Privileged access will require separate administrative accounts, and direct user-level permission assignments will be avoided. Regular access reviews will be conducted to maintain a strong security posture as the organisation grows.

Conditional Access Strategy

Conditional Access will be implemented as a core security control. Multi-Factor Authentication will be required for all users. Access to sensitive resources, such as finance records and user databases, will require compliant and managed devices.

High-risk sign-ins will be automatically blocked. Privileged roles will have additional restrictions, including access only from trusted locations and registered devices.

User consent for third-party applications will be restricted to prevent OAuth abuse and excessive permission grants. This layered security model reduces the risk of account compromise and data exposure.

Microsoft Entra ID Edition Selection

Microsoft Entra ID P1 is the recommended edition. The organisation handles sensitive information, including user databases and financial records, which require Conditional Access policies, Multi-Factor Authentication, and device compliance controls. P1 provides the necessary features while remaining cost-effective for a growing mid-sized company. If the company expands or regulatory requirements increase, upgrading to P2 would be considered for advanced identity protection and privileged access management.